Details of Coursework

The task for the coursework is that **you will divide yourselves into groups of 3 and pick a reference (related to the lectures but in topics not covered in them) to read and present in detail as a group**. **Everyone in the group gets the same score --- the poster coursework is a group project.** You will prepare and submit a poster and do an associated **15 minute poster presentation**, and prepare to answer questions based on the paper/poster.

The poster, the oral presentation and the Q&A session, all 3 will be graded.
The presentation material is your choice.

You might want to use generative-AI to help with the coursework,
but if you do so, you are necessarily required to declare its usage.

Note the set of instructions numbered A to H that follow.
And at the end of this page we have given a list of suggestions but you can pick your own - just check with us first.

A.
You are responsible for finding your own team, and notifying us of the members.
Notification deadline is 27th March i.e the last Friday before Easter Break by when you should let us know as to who is your group of 3 and what are you preparing the poster on. **So, you have ~2 months from the start of the course to decide on this outline plan.** One of your group members needs to send me (`anirbit.mukherjee@manchester.ac.uk`) an email with the details, with cc to other group members with an email subject title of **[COMP 34312 Poster Details]**.

B.
The poster session will be 10 AM - 12 PM and then 1PM - 2PM on 8th May with a reserve day of 11th May at the same times, if we cannot fit everyone on the first option. *You are advised to come before time and set up the poster*.

So, you have ~3 months from the start of the course to prepare your poster. This is not at all a long time to prepare given what is expected of a good poster **(see the point E)** and **hence, you are advised to read ahead and start preparing immediately.**

C.
Submission will require a PDF of the poster, apart from the in-person on a screen presentation explained above.

D.
The rooms Collab 1 and 2 have 12 setups of a large screen with a desk with several chairs. For their presentation, each group will be assigned a time slot and one of these setups. They would project their poster onto the screen and present it to the marker assigned to that session.

In any one round, there would be at most 6 groups presenting simultaneously at their assigned alternate setups spread across Collab 1 and 2. The markers will be the lecturers and some of the GTAs. We anticipate requiring not more than 6 rounds, each with at most 6 presentations and each round should take not more than 30 minutes.

E.
You are advised to carefully study the advice in the following file,
How-To-Score-High-On-Poster-Coursework.pdf (https://canvas.manchester.ac.uk/courses/42767/files/15201567?wrap=1)

F.
Software: you are free to use any software you wish to create the poster:
Powerpoint, Google Slides, Latex, are all good choices.

For inspiration here are some examples of posters from various ML academics.
Poster-Burgers.pdf, (https://canvas.manchester.ac.uk/courses/42767/files/15196689?wrap=1) Poster-LMC.pdf (https://canvas.manchester.ac.uk/courses/42767/files/15196653?wrap=1) ,Poster-Features.pdf (https://canvas.manchester.ac.uk/courses/42767/files/15196861?wrap=1) , Poster-MI.pdf, (https://canvas.manchester.ac.uk/courses/42767/files/15196852?wrap=1) Poster-Stability.pdf, (https://canvas.manchester.ac.uk/courses/42767/files/15196838?wrap=1) Poster-Information.pdf (https://canvas.manchester.ac.uk/courses/42767/files/15238382?wrap=1) (https://canvas.manchester.ac.uk/courses/42767/files/15196838?wrap=1)

You can find the official UoM poster template **here, (https://studentnet.cs.manchester.ac.uk/pgr/symposium/latextemplate.zip)**
-which is set to a portrait mode.

If you prefer a landscape mode then as a poster template you could use **this (https://github.com/Anirbit-AI/ICML_2028_Workshop_Poster)** .
This also acts another ML poster example.

Both of the above are a collection of LaTeX files which is best compiled as a single project on **Overleaf (http://overleaf.com/)** .

G.
Here are some links to advice on how to make a good poster,
https://deeplearningindaba.com/2022/indaba/posters/

(https://deeplearningindaba.com/2022/indaba/posters/)
https://www.animateyour.science/post/best-examples-of-scientific-posters (https://www.animateyour.science/post/best-examples-of-scientific-posters)
https://graphicdesign.stackexchange.com/questions/9452/examples-of-good-academic-poster-design (https://graphicdesign.stackexchange.com/questions/9452/examples-of-good-academic-poster-design)

H.
Options for the Poster Coursework

- Whenever a theorem/lemma number is mentioned as a suggestion its expected that its proof will be an integral part of the eventual presentation.

- Whenever a proof is presented, it is expected that proofs of all results invoked inside it are also to presented.

- The suggested references span a range of technical depth.
  To help you choose a topic we have done a 3-way labelling of the references,

- **[CORE]**
  These could easily have been covered in a few additional lecture sessions in this course.
- **[INTERMEDIATE]**
  They are related to the lecture topics and start to introduce slightly more involved ideas than what has been taught.
- **[CHALLENGING]**
  To understand these, you'll need to learn a significantly new concept beyond the lectures.

- Selecting a CHALLENGING reference will not guarantee higher grades, just as choosing a CORE marked option will not restrict what grades you can achieve. No reference is inherently better than another; they simply differ in focus.

The following paper suggestions are based on CORE material of the course.

Title	Details
Bias/Variance is not the same as Approximation/Estimation	https://openreview.net/pdf?id=4TnFbv16hK (https://openreview.net/pdf?id=4TnFbv16hK)
Reconciling modern machine learning	https://arxiv.org/abs/1812.11118 (https://arxiv.org/abs/1812.11118)

practice and the bias-variance trade-off	
KNN in high-dimensions	https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html (https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html) For more details feel free to check the paper linked at the bottom of the above and this, https://www.jmlr.org/papers/volume11/radovanovic10a/radovanovic10a.pdf (https://www.jmlr.org/papers/volume11/radovanovic10a/radovanovic10a.pdf)
Do a study of AdaBoost	https://engineering.purdue.edu/kak/Tutorials/AdaBoost.pdf (https://engineering.purdue.edu/kak/Tutorials/AdaBoost.pdf)
Do a study of MLE and MAP	You might want to start with this, https://www.robots.ox.ac.uk/~az/lectures/est/lect34.pdf (https://www.robots.ox.ac.uk/~az/lectures/est/lect34.pdf) Also see these set of 3 lectures, https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/MLE_MAP_Part1.pdf (https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/MLE_MAP_Part1.pdf) https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/MLE_MAP_Part2.pdf (https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/MLE_MAP_Part2.pdf) https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/BayesClassifier.pdf (https://www.cs.cmu.edu/~aarti/Class/10701_Spring14/slides/BayesClassifier.pdf) This might also be helpful, https://www.cs.ubc.ca/~fwood/CS340/lectures/L26.pdf (https://www.cs.ubc.ca/~fwood/CS340/lectures/L26.pdf)
Do a study of the various popular deep-learning algorithms.	Interestingly all good deep-learning algorithms are gradient based optimizers! https://www.ruder.io/optimizing-gradient-descent/ (https://www.ruder.io/optimizing-gradient-descent/)
Do a study on the interplay between learning-rate and batch-size.	You might want to start from here, https://arxiv.org/abs/1711.00489 (https://arxiv.org/abs/1711.00489) (if interested feel free to enquire for more modern references on the above!)
Do a study on the double-descent phenomenon in deep-learning.	You might want to start from here, https://openai.com/index/deep-double-descent/ (https://openai.com/index/deep-double-descent/) The detailed version of the above overview is in the following paper,

	https://iopscience.iop.org/article/10.1088/1742-5468/ac3a74 (https://iopscience.iop.org/article/10.1088/1742-5468/ac3a74)
Do a study on the idea of "Causal ML"	Try Chapters 9 and 10 from https://mlstory.org (https://mlstory.org)
Understanding deep-learning requires rethinking generalization.	https://arxiv.org/abs/1611.03530 (https://arxiv.org/abs/1611.03530)
Do a study of Gradient Descent (GD) convergence on smooth convex functions.	This is already suggested as background reading for Chapter-9 of the notes, Chapter-9 uses this result without proof. Theorem 3.4 in https://arxiv.org/pdf/2301.11235 (https://arxiv.org/pdf/2301.11235) This is a foundational result about GD. Note that 2 different proofs are given and there is a Corollary 3.5. Try to cover all of it.
What is a Good Step-Length for GD? (Level 1)	Lemma 5.3 in https://openreview.net/pdf?id=ABT1XQLbOx (https://openreview.net/pdf?id=ABT1XQLbOx) **(this was work done by a previous student of the class COMP34312)**
Fastest Gradient Based Optimization of Smooth Convex Functions (Level 1)	> *This is arguably the most famous result in the history of GD.* Nesterov Acceleration : as given in Theorem 3.19 of https://arxiv.org/pdf/1405.4980 (https://arxiv.org/pdf/1405.4980)
Momentum Gradient Descent	https://distill.pub/2017/momentum/ (https://distill.pub/2017/momentum/)
Sub-Gradient Descent	Theorem 4.2 (and maybe 4.3), here (https://yuxinchen2020.github.io/large_scale_optimization/lectures/subgradient_methods.pdf) .
SGD for Strongly Convex Functions	Theorem 11.1 and 11.2, https://yuxinchen2020.github.io/large_scale_optimization/lectures/stochastic_gradient.pdf (https://yuxinchen2020.github.io/large_scale_optimization/lectures/stochastic_gradient.pdf)
Do a study on the idea of "algorithmic stability"	You might want to start from Proposition 1 in Chapter 6 of https://mlstory.org (https://mlstory.org/pdf/generalization.pdf) To understand Proposition 1, one might need to read pages 1 to 13

The following suggestion can be said to be at an INTERMEDIATE level.

Title	Details
What is a Good Step-Length for GD? (Level 2)	Lemma 5.3 in **https://openreview.net/pdf?id=ABT1XQLbOx** **(https://openreview.net/pdf?id=ABT1XQLbOx)** **And** also want to cover Theorem 2.2 **(this was work done by a previous student of the class COMP34312)**
Fastest Gradient Based Optimization of Smooth Convex Functions (Level 2)	> *This is arguably the most famous result in the history of GD.* Nesterov Acceleration : as given in Theorem 3.19 of **https://arxiv.org/pdf/1405.4980** **(https://arxiv.org/pdf/1405.4980)** **And** also cover the argument for why the above is the fastest possible such method, as explained in Theorem 3.14 of above. You can also see other such speed-limits given in Theorems 3.13 and 3.15 of above
Interpolation, PL Condition and Exponentially Fast SGD	Theorem 1 in **https://arxiv.org/pdf/1811.02564** **(https://arxiv.org/pdf/1811.02564)** (optionally also cover Theorem 2)
VC Dimension Upperbounds Rademacher Complexity - for Binary Class Predictors.	Page-1 of, **https://home.ttic.edu/~tewari/lectures/lecture10.pdf** **(https://home.ttic.edu/~tewari/lectures/lecture10.pdf)** Then, **https://home.ttic.edu/~tewari/lectures/lecture11.pdf** **(https://home.ttic.edu/~tewari/lectures/lecture11.pdf)**
Do a study of the basics of formal theory of optimization.	See Chapters 2 and 3 here, **https://damek.github.io/teaching/orie6300/ORIE6300Fall2023notes.pdf** **(https://damek.github.io/teaching/orie6300/ORIE6300Fall2023notes.pdf)** Try to do (and present) some of the exercises - even if the easier ones.

These papers are somewhat more CHALLENGING.
They go well beyond the scope of the course, but you may like to take on a challenge...

Title	Details

What is a Good Step-Length for GD? (Level 3)	Lemma 5.3 in **https://openreview.net/pdf?id=ABT1XQLbOx** (https://openreview.net/pdf?id=ABT1XQLbOx) **And** Theorem 2.1 **(this was work done by a previous student of the class COMP34312)**
Fastest Gradient Based Optimization of Smooth Convex Functions (Level 3)	*> This is arguably the most famous result in the history of GD.* Nesterov Acceleration : as given in Theorem 3.19 of **https://arxiv.org/pdf/1405.4980** (https://arxiv.org/pdf/1405.4980) **And** somewhat surprisingly, very recently ChatGPT could be used to discover new things about Nesterov Acceleration algorithm. See, **https://arxiv.org/abs/2510.23513** (https://arxiv.org/abs/2510.23513)
Low Degree Algorithm for PAC Learning	Theorem 3.29 in **https://arxiv.org/pdf/2105.10386** (https://arxiv.org/pdf/2105.10386) and its specific consequence given on page 81. To understand this, it is possibly necessary to read Section 3.4 from the start, and a few things before that.
What is a Good Step-Length for (Stochastic) GD?	There is no clear answer to this, but see, Theorem 2 of **https://arxiv.org/pdf/1912.03194** (https://arxiv.org/pdf/1912.03194) To appreciate this notice the failure case of basic SGD explained in Remark-1. You might also consider covering Theorem 3 and the much advanced Section 3.2
Linear Regression Over Features and Very Wide Neural Nets	**https://rbcborealis.com/research-blogs/the-neural-tangent-kernel/** (https://rbcborealis.com/research-blogs/the-neural-tangent-kernel/)
Gradient Flow	**https://rbcborealis.com/research-blogs/gradient-flow/** (https://rbcborealis.com/research-blogs/gradient-flow/)
A General Upperbound on Rademacher Complexity	Sections 1 and 2 of **https://home.ttic.edu/~tewari/lectures/lecture14.pdf** (https://home.ttic.edu/~tewari/lectures/lecture14.pdf) Sections 1 and 2 of

	https://home.ttic.edu/~tewari/lectures/lecture15.pdf (https://home.ttic.edu/~tewari/lectures/lecture15.pdf)